

AI2: Bayesian Networks – Representation

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Outline

We learn about:

- ▶ Curse of dimensionality
- ▶ Directed acyclic graphical model (DAG), also known as Bayes Networks (BN)
- ▶ Conditional independence structures in DAGs
- ▶ Application domains of Bayes nets

The curse of dimensionality

This curse tells us that to represent a joint distribution of d binary variables, we need $2^d - 1$ terms.

Example with $d = 3$:

X	Y	Z	Joint Probability
0	0	0	$P(X = 0, Y = 0, Z = 0)$
0	0	1	$P(X = 0, Y = 0, Z = 1)$
0	1	0	$\vdots$
0	1	1	$\vdots$
1	0	0	$\vdots$
1	0	1	$\vdots$
1	1	0	$\vdots$
1	1	1	$\vdots$

⇒ Total of terms.

A simple probabilistic graphical model

Let us revisit our problem of drawing balls from a jar $\{r, r, r, b\}$, but this time *without replacement*.

$$P(B_1) = \frac{r \quad b}{\quad}$$

$$P(B_2 \mid B_1) = \frac{B_1 \setminus B_2 \mid r \quad b}{r \quad b}$$

$$P(B_1, B_2) = P(B_1)P(B_2 \mid B_1) = \frac{B_1 \setminus B_2 \mid r \quad b}{r \quad b}$$



A simple probabilistic graphical model

Let us revisit our problem of drawing balls from a jar $\{r, r, r, b\}$, but this time *without replacement*.

$$P(B_1) = \begin{array}{c|cc} & r & b \\ \hline & \frac{3}{4} & \frac{1}{4} \end{array}$$

$$P(B_2 | B_1) = \begin{array}{c|cc} B_1 \backslash B_2 & r & b \\ \hline r & \frac{2}{3} & \frac{1}{3} \\ b & 1 & 0 \end{array}$$

$$P(B_1, B_2) = P(B_1)P(B_2 | B_1) = \begin{array}{c|cc} B_1 \backslash B_2 & r & b \\ \hline r & \frac{1}{2} & \frac{1}{4} \\ b & \frac{1}{4} & 0 \end{array}$$

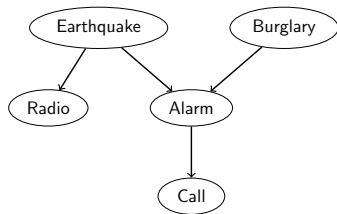


Directed probabilistic graphical models

1. Directed Acyclic Graph (DAG)

- ▶ **Nodes** – random variables
- ▶ **Edges** – direct influence (“causation”)

2. X_i independent of $X_{\text{ancestors}} \mid X_{\text{parents}}$



$$P(B, E, A, R, C)$$

$$= P(B)P(E \mid B)P(A \mid B, E)P(R \mid A, B, E)P(C \mid R, A, B, E)$$

$$= P(B)P(E)P(A \mid B, E)P(R \mid E)P(C \mid A)$$

The DAG tells us that if we have n variables x_i , the joint distribution of these variables *factorises* as follows:

$$P(x_{1:n}) = P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{parents}(x_i))$$

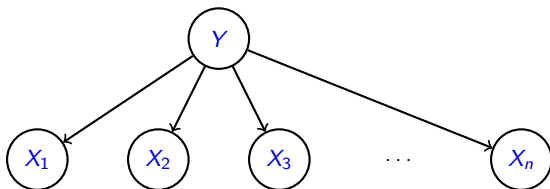
Example 1: Naïve Bayes as a Bayesian Network

The simplest Bayesian network is the Naïve Bayes Classifier:

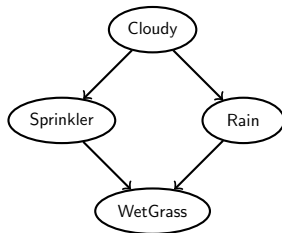
- ▶ Target variable Y (Class label).
- ▶ Input variables $X = \{X_1, X_2, \dots, X_n\}$ (Evidence).

Assumption encoded in the graph below:

- ▶ The input variables (data features) are conditionally independent given $Y = y$.



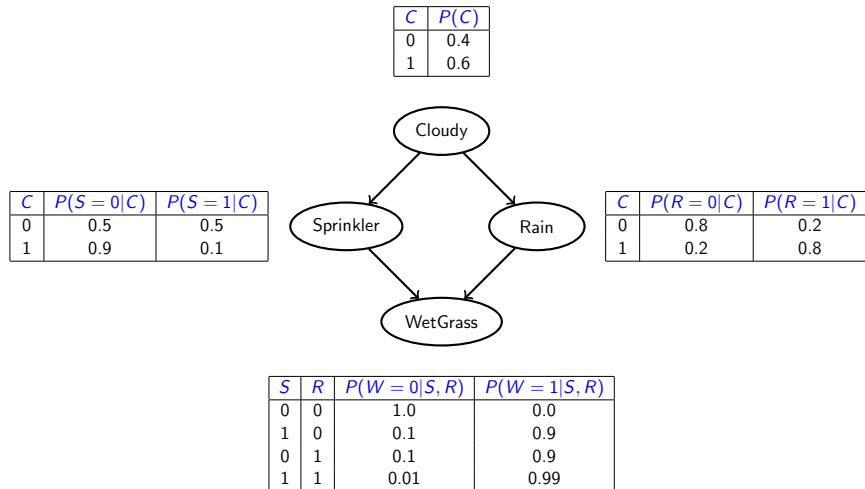
Another example: Wet grass



Discrete Random Variables:

- Conditional distribution can be represented as a conditional probability table (CPT) – the distribution over X_i for each combination of parent values.

CPTs in the Wet Grass Example



Q: How many parameters did this Bayes net model save us?

The formal meaning of a Bayesian Network

Instead of the **full joint distribution**:

$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2 | X_1)P(X_3 | X_1, X_2) \dots P(X_n | X_1, X_2, \dots, X_{n-1}),$$

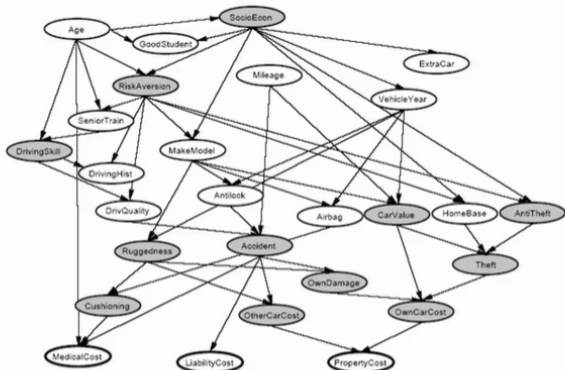
a Bayesian Network, encodes the following **limited-dependence assumption**:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i))$$

Essence of a Bayesian Network:

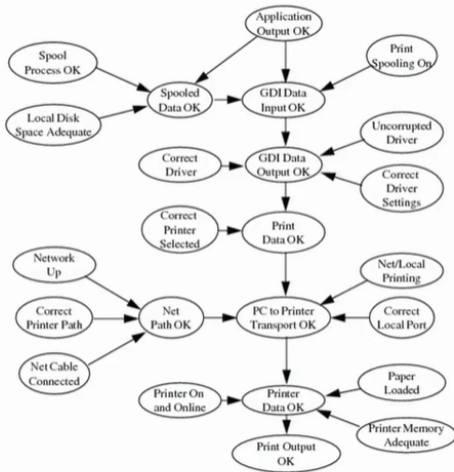
- ▶ A compact representation of a joint probability distribution in terms of conditional distributions.

Example: Vehicle insurance

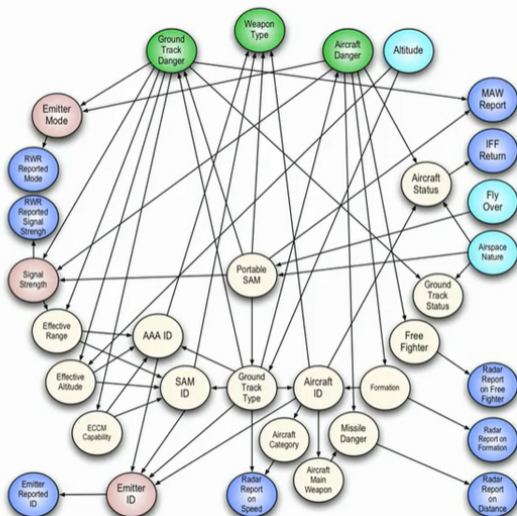


The 12 **shaded variables** are considered **hidden** or **unobservable**, while the other 15 are **observable**. The network has over 1400 parameters. An insurance company would be interested in predicting the bottom three "cost" variables, given all or a subset of the other observable variables.

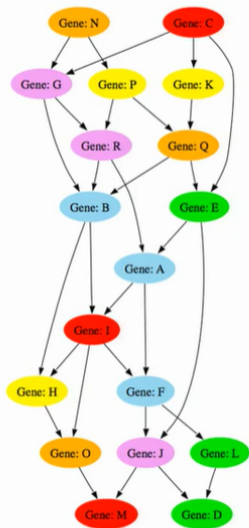
Example: Microsoft Windows Printer Troubleshooter



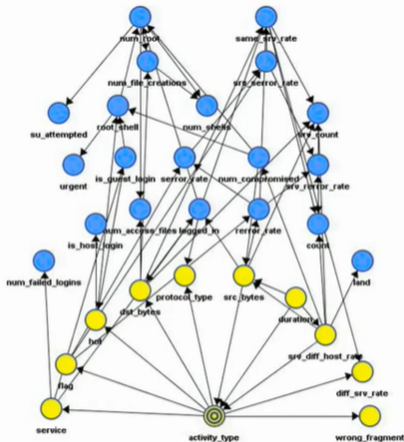
Example: Radar and aircraft control



Example: Gene expression

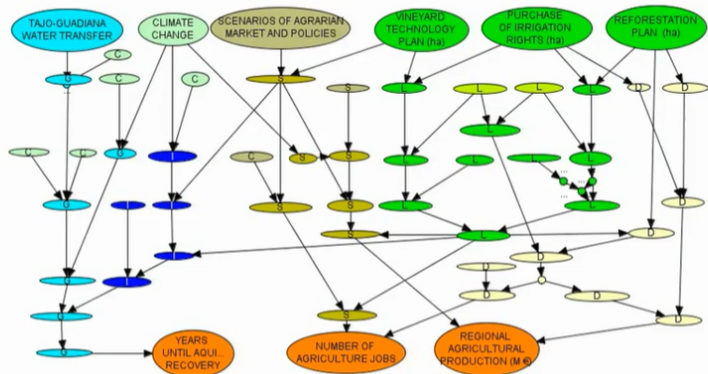


Example: Cyber crimes detection



<http://www.bayesia.com/en/products/bayesialab/resources/case-studies/cyber-crime-detection.php>

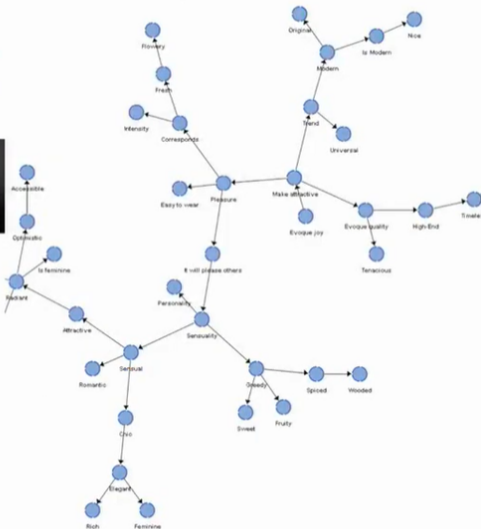
Example: Water resource management



Bayesian network of the Upper Guadiana basin at aquifer-scale (Zorrilla 2009).

“G and light blue” variables refer to **groundwater**; “I and blue” represent **irrigation variables**; “C and light green” variables correspond to **climate**; “S and brown” variables represent **socio-economic scenarios**; “L and green” variables relate to **irrigated land**; and “D and yellow” variables represent **rain-fed agriculture**.

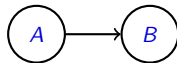
Example: Marketing & fashion



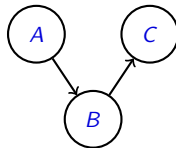
<http://www.bayesia.com/en/applications/marketing.php>

Probabilistic Relationships: Standard Structures

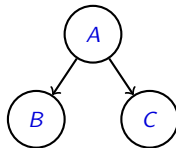
a) Direct Influence



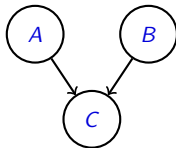
b) Indirect Influence (chain)



c) Common Cause

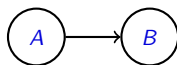


d) Common Effect



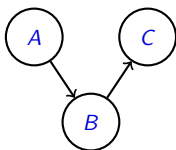
Probabilistic Structures & Conditional Independences

a) Direct Influence



$$P(A)P(B | A)$$

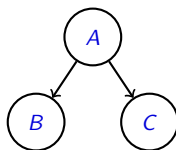
b) Indirect Influence (chain)



$$P(A)P(B | A)P(C | A)$$

$$A \perp\!\!\!\perp C | B$$

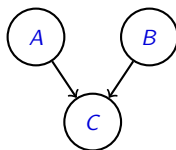
c) Common Cause



$$P(A)P(B | A)P(C | B)$$

$$B \perp\!\!\!\perp C | A$$

d) Common Effect



$$P(A)P(B)P(C | A, B)$$

$$A \not\perp\!\!\!\perp B | C$$

Conditional Independence

Recall: Two sets of variables A and B are **independent** if:

$$P(A) = P(A | B) \quad \text{or equivalently} \quad P(A, B) = P(A)P(B)$$

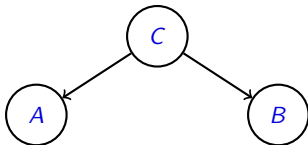
Graphical Representation of Independence:



Conditional Independence: Two random variables A and B are conditionally independent given a third random variable C , written as:

$$(A \perp\!\!\!\perp B) | C \quad \text{if and only if} \quad P(A, B | C) = P(A | C)P(B | C)$$

Graphical Representation of Conditional Independence:

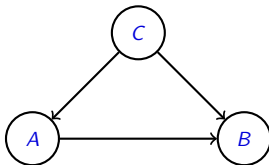


Conditional Independence: Why It Matters?

The Full Joint Distribution:

$$P(A, B, C) = P(C)P(A | C)P(B | A, C)$$

Graphical Representation:



Parameter Count:

- ▶ Suppose A , B , and C are binary variables (0 or 1).
- ▶ Full joint distribution requires $1 + 2 + 4 = 7$ parameters.
- ▶ With conditional independence, as on the previous slide, we have $P(A, B, C) = P(C)P(A | C)P(B | C)$ so only need $1 + 2 + 2 = 5$ parameters.

(Local) Markov property

The Bayes Net encodes a **(local) Markov condition**: Each random variable X is conditionally independent of its non-descendants, given its parents. That is, $X \perp \text{non-descendants} \mid \text{parents}$.

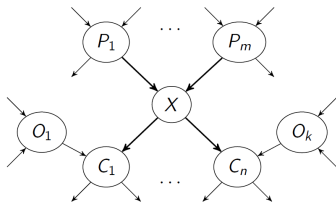
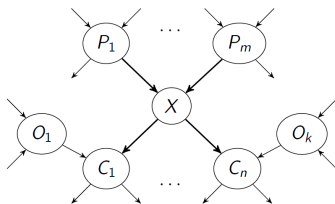


Figure: $P_1, \dots, P_m$ are **parents** of node X . $C_1, \dots, C_n$ are **descendants or children** of X . All other nodes ($O_1, \dots, O_k$) are neither parents nor children of X . Hence, X is conditionally independent of all $O_1, \dots, O_k$ given $P_1, \dots, P_m$. That is, $X \perp\!\!\!\perp O_1, \dots, O_k \mid P_1, \dots, P_m$.

Guarantees that X is independent of nodes that aren't affected by it (non-descendants), once the parent nodes' values are known.

"I only need to know my parents to ignore my non-descendants."

Markov Blanket



Definition: For node X , the Markov blanket is:

- ▶ Parents of X : $P_1, \dots, P_m$
- ▶ Children of X : $C_1, \dots, C_n$
- ▶ Co-parents of X 's children: $O_1, \dots, O_k$

The Markov blanket is sufficient for inferring X . That is, all information about a random variable (X) is contained in its Markov blanket.

$X \perp \text{Other nodes} \mid \text{Markov blanket of } X$

"I only need to know my Markov blanket (parents, children, and co-parents) to ignore *everything else* in the network."

D-Separation (Conditional Independence)

D-separation is a graphical criterion that lets you determine *all conditional independencies* implied by the network (not just independence from ancestors given parents).

Definition: Two nodes A and C are **d-separated** by a set of observed nodes Z if all paths between them are **blocked** by Z . **This means:**

$$A \perp\!\!\!\perp C \mid Z$$

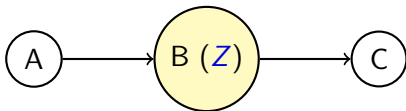
Blocking rules:

- ▶ **Chain / Fork:** $A \rightarrow B \rightarrow C$ or $A \leftarrow B \rightarrow C \rightarrow$ blocked if the middle node $\in Z$
- ▶ **Collider:** $A \rightarrow B \leftarrow C \rightarrow$ blocked if the middle node $\notin Z$ and none of its descendants $\in Z$

D-Separation Examples

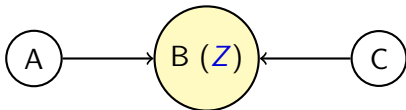
1. Chain: $A \rightarrow B \rightarrow C$

- ▶ Without conditioning ($Z = \emptyset$): $A \not\perp C$ (dependent)
- ▶ Conditioning on B ($Z = \{B\}$): $A \perp C \mid B$ (conditionally independent)



2. Collider: $A \rightarrow B \leftarrow C$

- ▶ Without conditioning ($Z = \emptyset$): $A \perp C$ (independent)
- ▶ Conditioning on B ($Z = \{B\}$): $A \not\perp C \mid B$ (conditionally dependent)



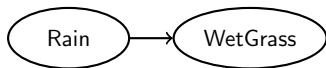
Markov Blanket & D-Separation Connection

- ▶ The Markov blanket guarantees that X is **d-separated** from all other nodes outside the blanket.
- ▶ Practical use:
 - ▶ Feature selection in ML
 - ▶ Simplified inference in Bayesian networks
 - ▶ Causal reasoning

Before concluding, in the remaining slides, we pick apart the 'Wet Grass' example and identify the found structures (3 conditional dependence structures) that we talked about on slides 18-19.

a) Direct Influence

Example:



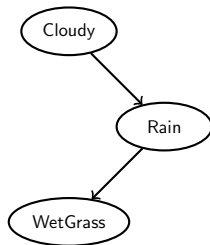
We can use Binary Random Variables:

- ▶ $W \in \{0, 1\}$ represents whether the grass is wet (1) or not (0).
- ▶ $R \in \{0, 1\}$, represents whether it has been raining (1) or not (0).

Probabilistic relationship: An edge represents the conditional dependence (influence, or "cause-effect" relationship) between the parent node (cause) and the child node (effect): $P(W|R)$.

Probabilistic relationships: b) Indirect influence

Consider the following: Cloudy influences Rain, so we have:



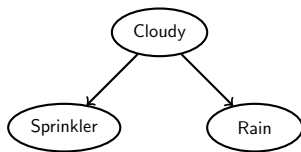
Hence, **WetGrass** is independent of **Cloudy** given **Rain** .

The joint distribution encoded in this graph is:

$$P(C, R, W) = P(C)P(R|C)P(W|R).$$

Probabilistic relationships: c) Common cause

It is natural to suppose the cloudy weather influence our decision to turn on the sprinkler or not and cloudy weather will also influence the chance of rain, we model this “common cause” probabilistic relationship using:



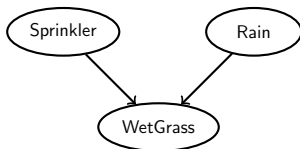
The joint probability distribution:

$$P(C, S, R) = P(C)P(S|C)P(R|C)$$

If C is observed, then S and R are independent.

Probabilistic relationships: d) Common effect

Suppose we identify another cause of wet grass, the sprinkler, we can then model the relationship as a converging connection (Common effects):



The joint probability distribution:

$$P(S, R, W) = P(S)P(R)P(W|S, R)$$

If neither W nor any of its descendants are observed, then S and R are independent.

Explaining away: If we can confirm one cause (e.g., rain is true) of an observation (e.g., grass is wet), it reduces the need to invoke alternative causes, e.g., sprinkler is on.

Summing up - Bayesian Networks Pros and Cons

Advantages:

- ▶ Graphical representation: Provides a visual representation of joint probability distributions of different random variables – interpretability.
- ▶ Powerful: Can capture complex relationships between random variables.
- ▶ Combine data and prior knowledge: Incorporates prior knowledge and updates with statistically significant information from data.
- ▶ Generative approach: Able to generate new data similar to existing data.

Disadvantages:

- ▶ Requires prior knowledge of many probabilities.
- ▶ Sometimes computationally intractable.

Further readings

- ▶ Stuart J. Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, Chapter 14.
- ▶ Poole & Mackworth: Artificial Intelligence: Foundations of Computational Agents, 3rd edition, Chapter 9
- ▶ M. Jordan: Graphical Models
- ▶ Koller & Friedman: Probabilistic Graphical Models: Chapters 3.1-3.3 (you may find the online version yourself.)
- ▶ Smyth: Note set 2 - Conditional independence and graphical models